

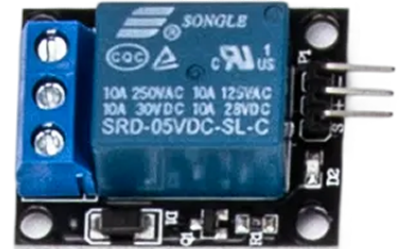
3.5 relay

Relay

In this tutorial, you will learn how to use a relay.

Component Required:

- (1)x Elegoo ESP32
- (2) x 400 tie-points breadboard
- (1) x Fan blade and 3–6v dc motor
- (1) x L293D IC
- (1) x 5v Relay
- (1) x Power Supply Module
- (1) x 9VIA Adapter
- (15) x M-M wires (Male to Male jumper wires)



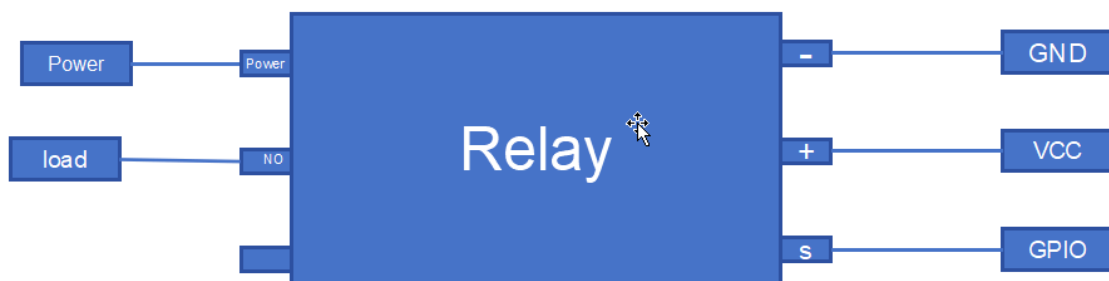
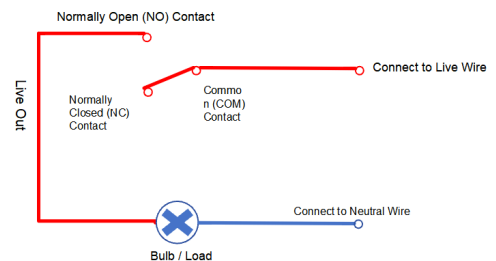
Component Introduction

Relay:

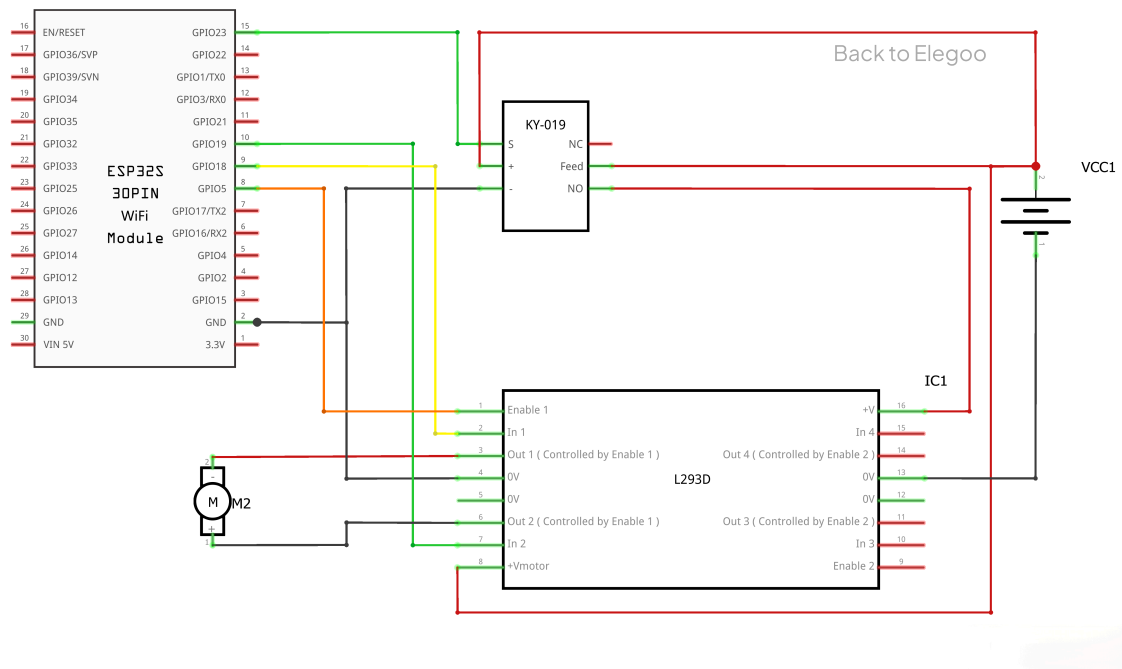
A relay is an electrically operated switch. Many relays use an electromagnet to mechanically operate a switch, but other operating principles are also used, as in solid-state relays. Relays are used where it is necessary to control a circuit by a low-power signal (with complete electrical isolation between control and controlled circuits), or where several circuits must be controlled by one signal. The first relays were used in long-distance telegraph circuits as amplifiers. They repeated the signal coming in from one circuit and re-transmitted it on another circuit. Relays were used extensively in telephone exchanges and early computers to perform logical operations.

A type of relay that can handle the high power required to directly control an electric motor or other loads is called a contactor. Solid-state relays control power circuits with no moving parts, instead using a semiconductor device to perform the switching. Relays with calibrated operating characteristics are sometimes multiple operating coils, which are used to protect electrical circuits from overloads and faults. In modern electrical power systems, these functions are performed by digital instruments called 'protective relays'.

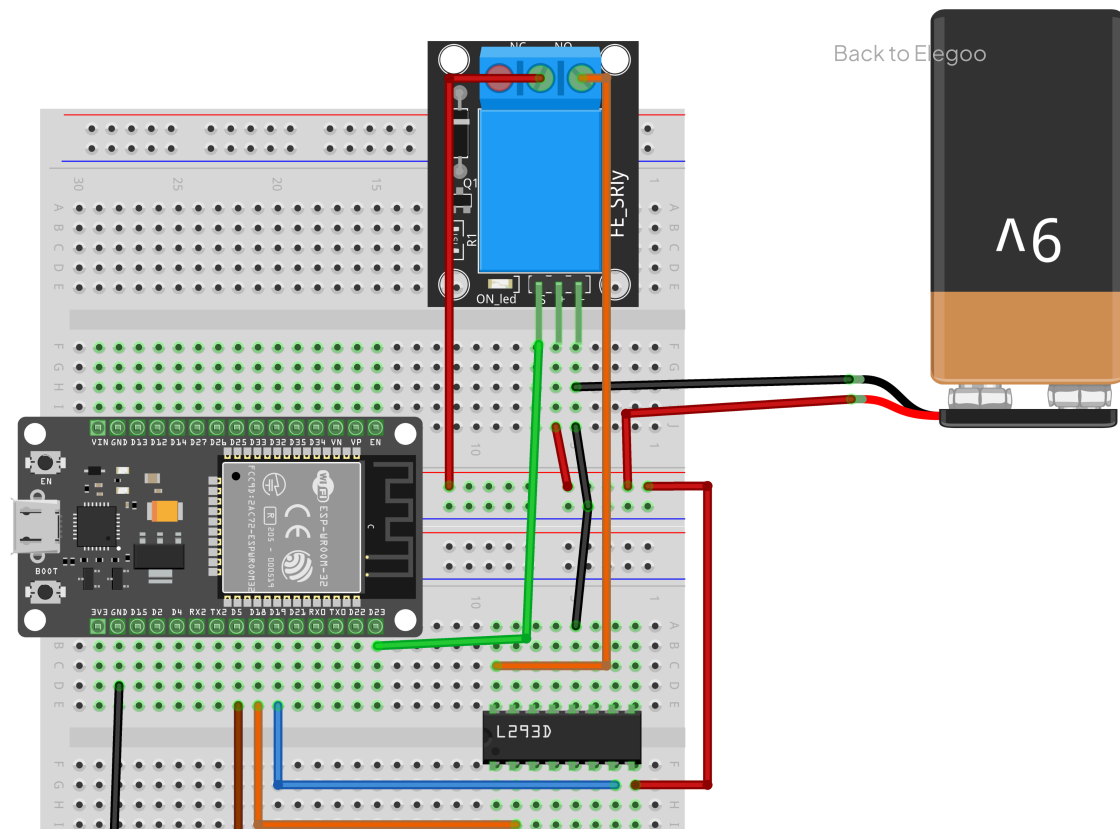
The figure below shows the pin diagram and wiring method of this relay: After the control terminal of the relay is connected to GPIO, GND and VCC, the relay will perform pull-in/release actions correspondingly when the GPIO outputs high/low level, and the bouncing sound of the internal spring can be heard during the process; if the power supply and wiring of the load terminal are normal at this time, the load will start and shut down with the action of the relay.



Connection Schematic



Wiring diagram



Motor Driver Pin Definitions:

- `#define RELAY_PIN 23` : Pin connected to relay

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```
void setup() {  
  //---set pin direction  
  pinMode(ENABLE,OUTPUT);  
  pinMode(DIRA,OUTPUT);  
  pinMode(DIRB,OUTPUT);  
  pinMode(RELAY_PIN,OUTPUT);  
  Serial.begin(9600);  
}
```

Setup Function:

- `pinMode(ENABLE,OUTPUT);` : Sets ENABLE pin as output
- `pinMode(DIRA,OUTPUT);` : Sets DIRA pin as output
- `pinMode(DIRB,OUTPUT);` : Sets DIRB pin as output
- `Serial.begin(9600);` : Initializes serial communication at 9600 baud rate for debugging

```
void loop() {  
  
  //---back and forth example  
  digitalWrite(RELAY_PIN,HIGH);  
  Serial.println("Relay ON: L293D powered up");  
  Serial.println("One way, then reverse");  
  digitalWrite(ENABLE,HIGH); // enable on  
  for (i=0;i<5;i++){  
    digitalWrite(DIRA,HIGH); //one way  
    digitalWrite(DIRB,LOW);  
    delay(750);  
    digitalWrite(DIRA,LOW); //reverse  
    digitalWrite(DIRB,HIGH);  
    delay(750);  
  }  
  digitalWrite(ENABLE,LOW); // disable  
  delay(3000);  
  for (i=0;i<5;i++){  
    digitalWrite(DIRA,HIGH); //one way  
    digitalWrite(DIRB,LOW);  
    delay(750);  
    digitalWrite(DIRA,LOW); //reverse  
    digitalWrite(DIRB,HIGH);  
    delay(750);  
  }  
  digitalWrite(ENABLE,LOW); // disable  
  digitalWrite(RELAY_PIN,LOW);  
  Serial.println("Relay OFF: L293D power cut off");  
  delay(3000);  
}
```

Loop Function:

- `digitalWrite(RELAY_PIN,HIGH);` Relay Pull-in
- `Serial.println("One way, then reverse");` : Prints status message to serial monitor
- `digitalWrite(ENABLE,HIGH);` : Enables the motor driver
- First `for` loop (5 iterations):
 - `digitalWrite(DIRA,HIGH); digitalWrite(DIRB,LOW);` : Sets motor direction forward
 - `delay(750);` : Runs motor forward for 750 milliseconds
 - `digitalWrite(DIRA,LOW); digitalWrite(DIRB,HIGH);` : Reverses motor direction
 - `delay(750);` : Runs motor reverse for 750 milliseconds
- `digitalWrite(ENABLE,LOW);` : Disables the motor driver
- `delay(3000);` : Pauses for 3 seconds
- Second `for` loop (5 iterations):
 - Same as first loop, repeats the forward/reverse cycle 5 times

- `digitalWrite(ENABLE,LOW);` : Disables the motor driver
- `delay(3000);` : Pauses for 3 seconds before repeating the loop

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Troubleshooting

- **Motor Not Moving:** Check all wiring connections and ensure L293D is properly powered
- **Motor Only Spinning One Direction:** Verify DIRA and DIRB connections to the motor driver
- **Motor Spinning Erratically:** Ensure proper grounding and stable power supply
- **No Serial Output:** Check serial monitor baud rate is set to 9600
- **Overheating L293D:** Ensure motor current does not exceed L293D's rating (600mA per channel)

[← Four digit segment display](#)

[Stepper motor with remote→](#)